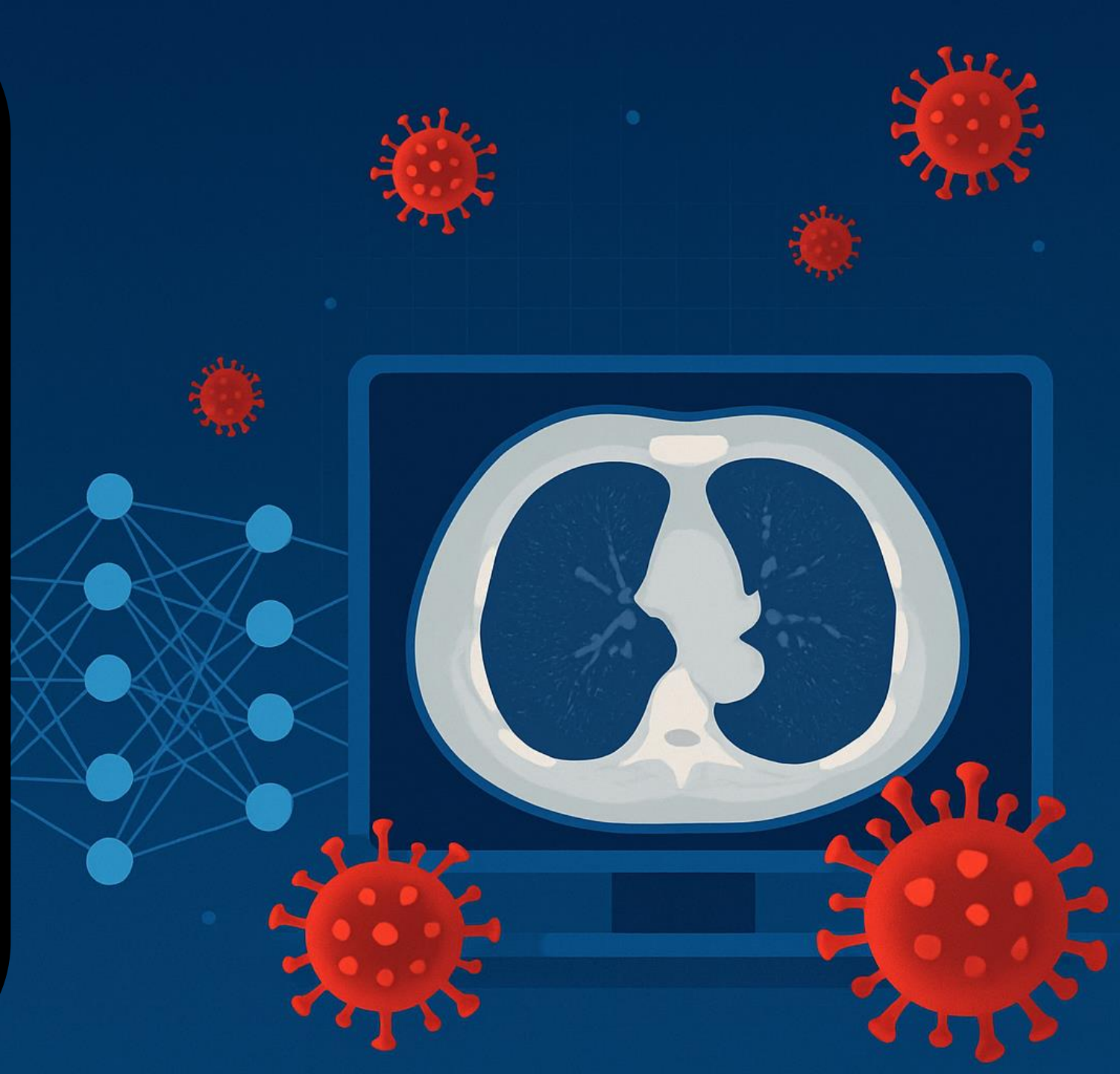


Deep Learning for CT-Based SARS-CoV-2 Detection and Classification

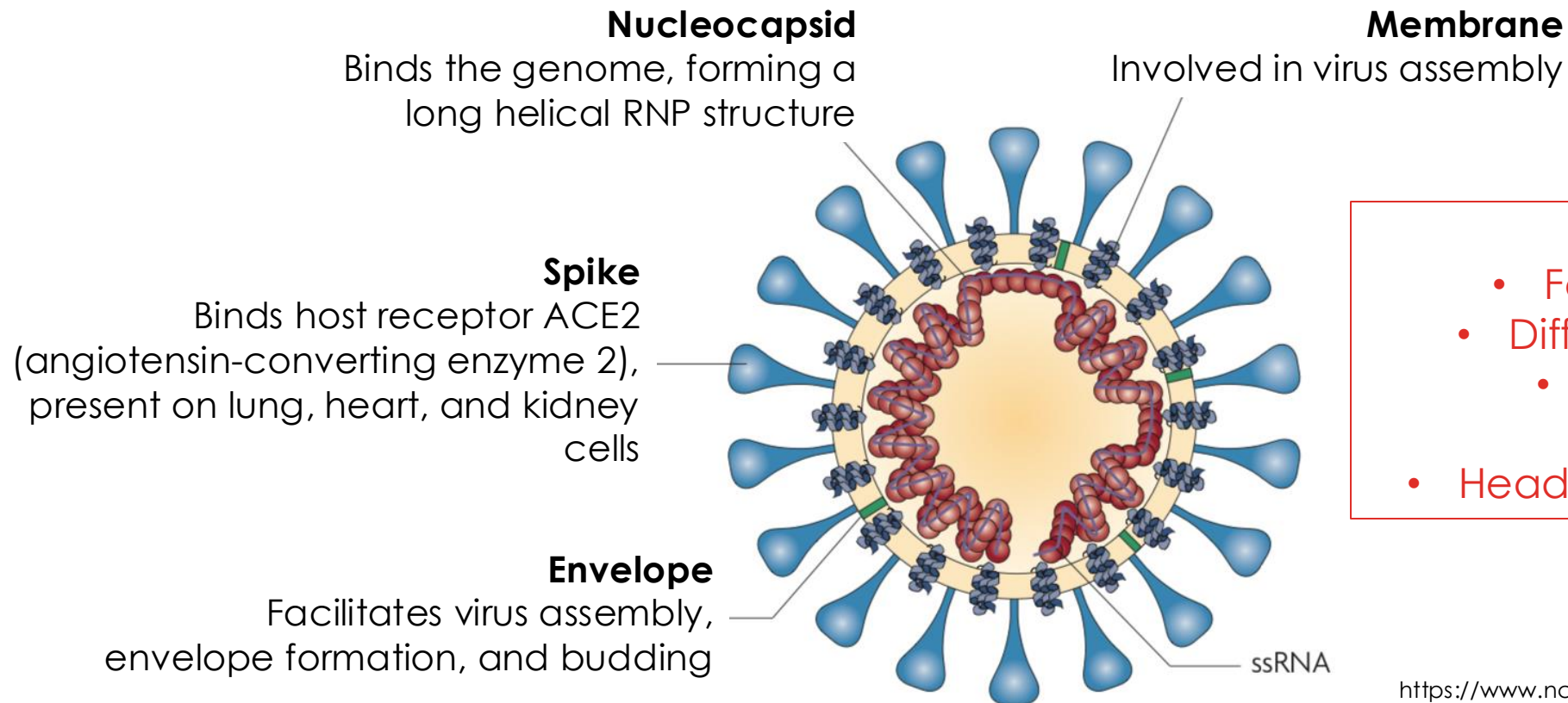
Presented by: Zayn Khan, Nitin
Sharan, Hanchuan Xu, Radchanon
(Dew) Leelasukseree



Department of Mathematics,
University of Wisconsin in Madison,
Madison, WI, USA



- SARS-CoV-2 (Severe Acute Respiratory Syndrome Coronavirus 2)
- COVID-19 (Corona Virus Disease 2019)

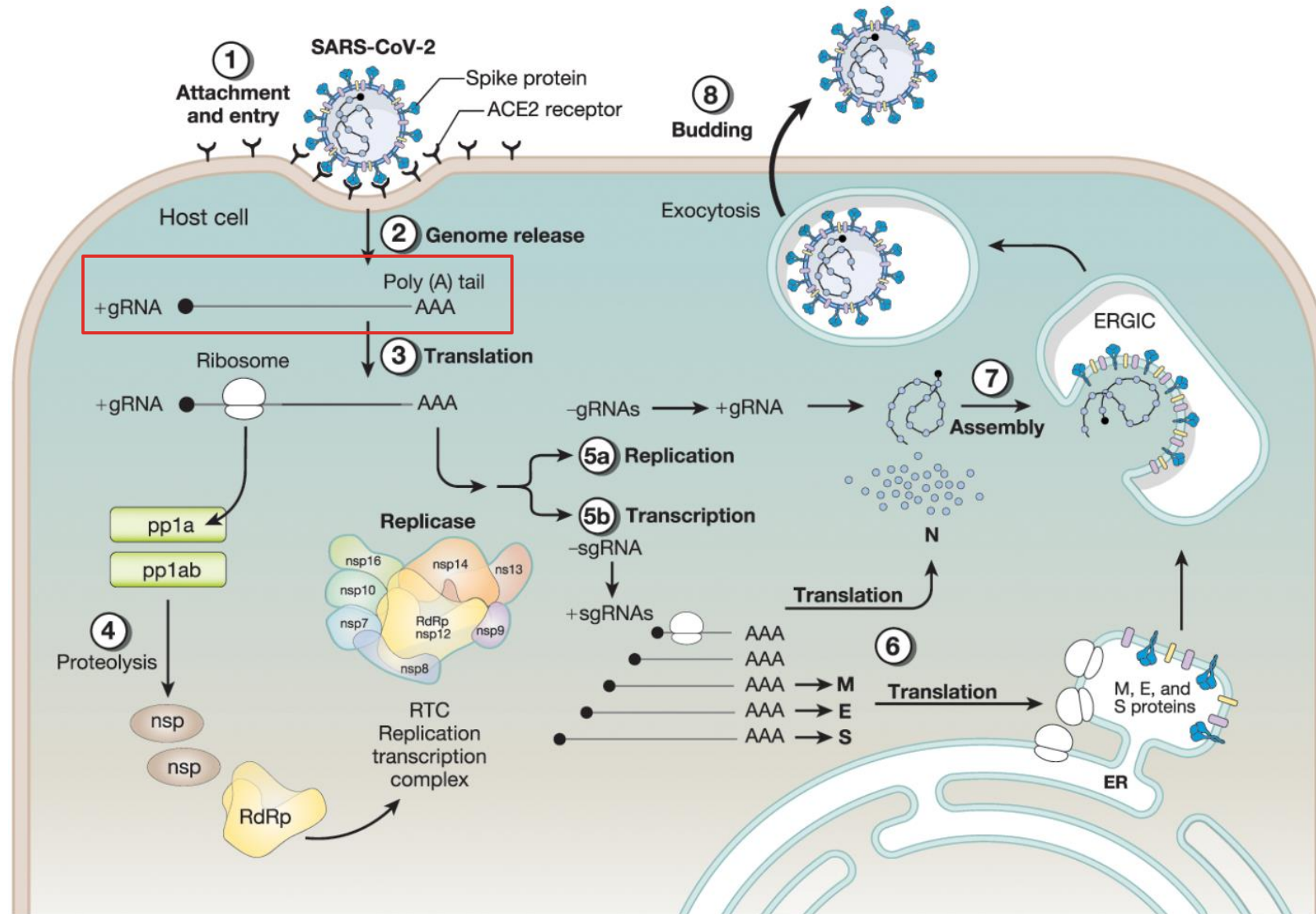


Symptoms

- Fever and chills
- Difficulty breathing
 - Sore throat
 - Fatigue
- Head and muscle ache

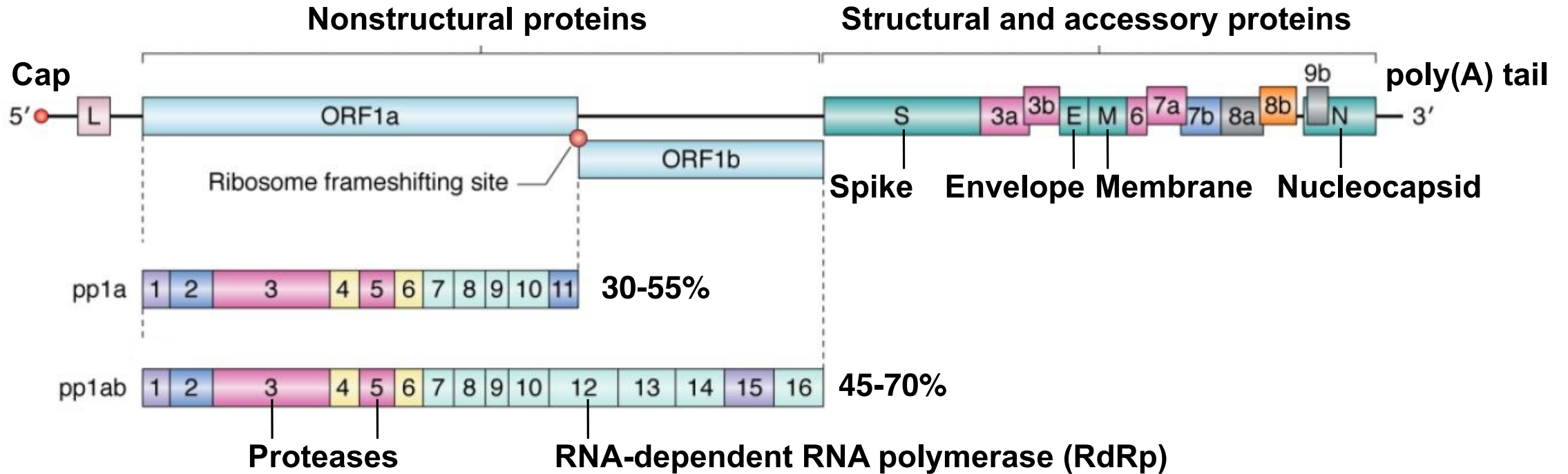
<https://www.ncbi.nlm.nih.gov/books/NBK554776/>

Perlman and Netland (2009) Nat Rev Microbiol 7, 439



Introduction

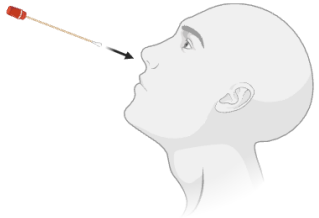
SARS-CoV-2 (+)-genomic RNA (+gRNA) schematic map



Hartenian et al. (2020) J. Biol. Chem.

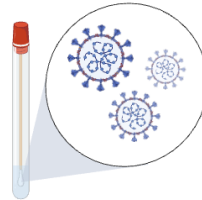
1 Nasopharyngeal swab <15 min

Cotton swab is inserted into nostril to absorb secretions.



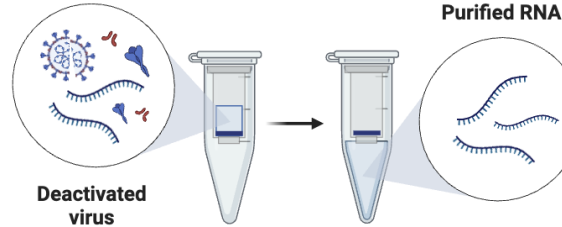
2 Collected specimen 0-72 h

Specimen is stored at 2-8°C for up to 72 hours or proceed to RNA extraction.



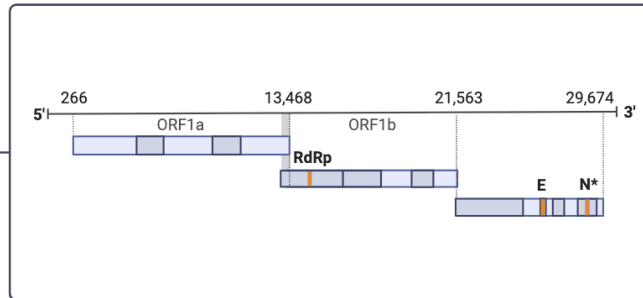
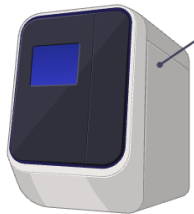
3 RNA extraction ~45 min

Purified RNA is extracted from deactivated virus.



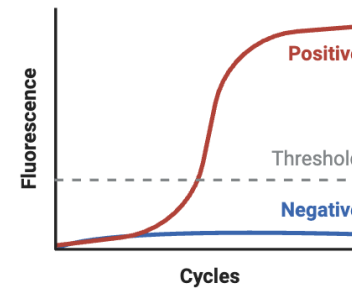
4 RT-qPCR ~1 h per primer set

Purified RNA is reverse transcribed to cDNA and amplified by qPCR.



5 Test results real-time

Positive SARS-CoV-2 patients cross the threshold line within 40.00 cycles (< 40.00 Ct).



Downfall of RT-qPCR

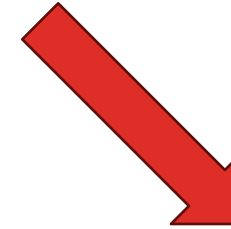
- Time-consuming
- Dependent on abundance of viral content in patient sample
 - Poor early stage diagnosis
- Expensive
- Absence of disease severity analysis (i.e. conditions of the lungs)

The Big Question

How can we enhance the speed and accuracy of early COVID-19 diagnosis while also providing disease severity assessments?

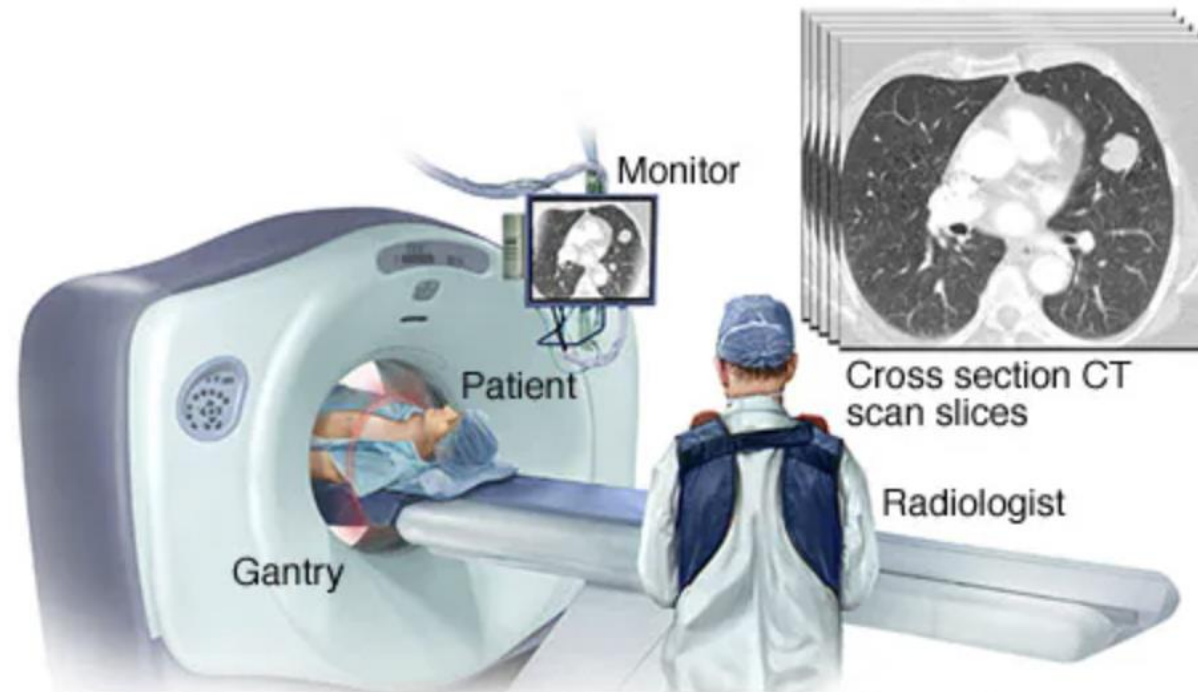


Patient computed tomography (CT) lung scans



Deep Learning Detection and Classification Model and Methods





© MAYO FOUNDATION FOR MEDICAL EDUCATION AND RESEARCH. ALL RIGHTS RESERVED.

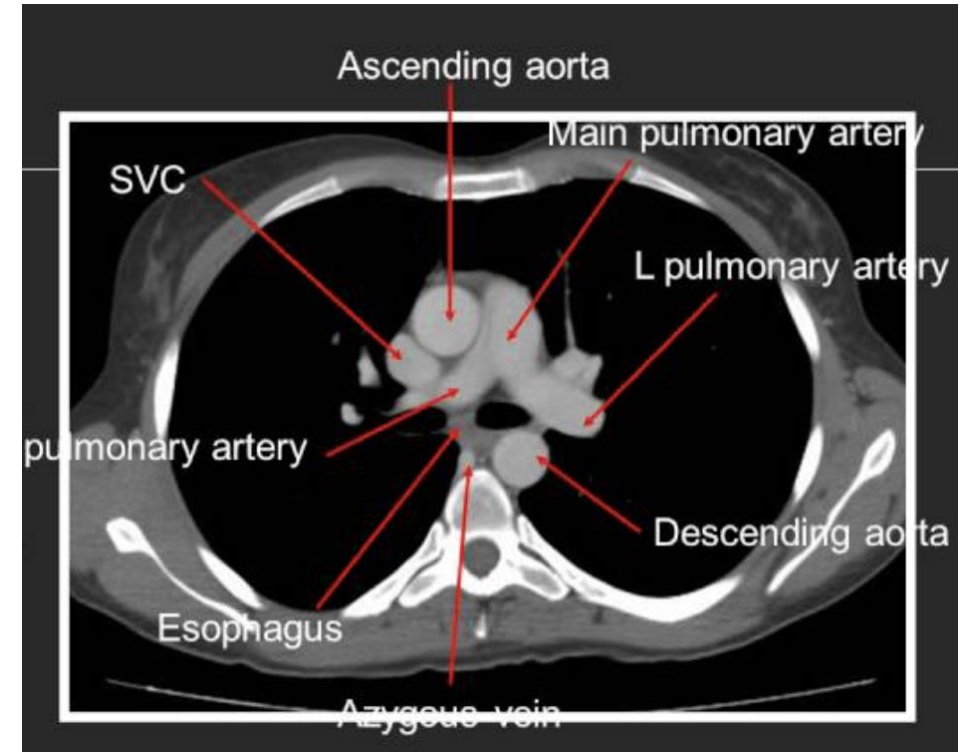
Procedure:

- Metals are removed
- Patients lie on their back on the “patient table”
- Patients are exposed to low levels of radiation for half-hour in the Gantry (even though scans take a few minutes)
 - Patients lie very still and hold their breath

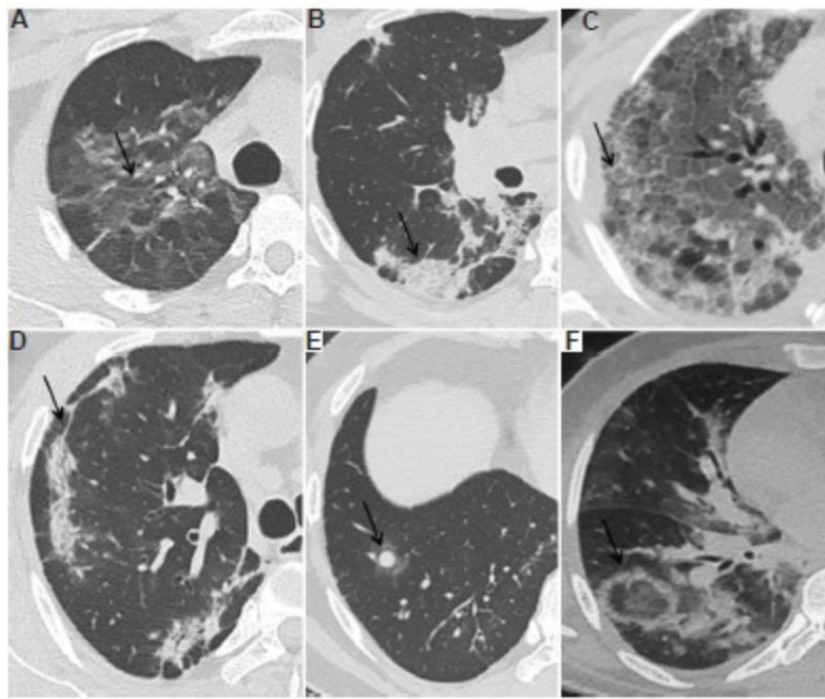
<https://www.mayoclinic.org/tests-procedures/lung-cancer-screening/about/pac-20385024>

Introduction

Transverse Computed Axial Tomography of the Thorax



https://www.youtube.com/watch?v=aE8tfmqld_Q
<https://www.imaios.com/en/e-anatomy/thorax/ct-axial-chest>



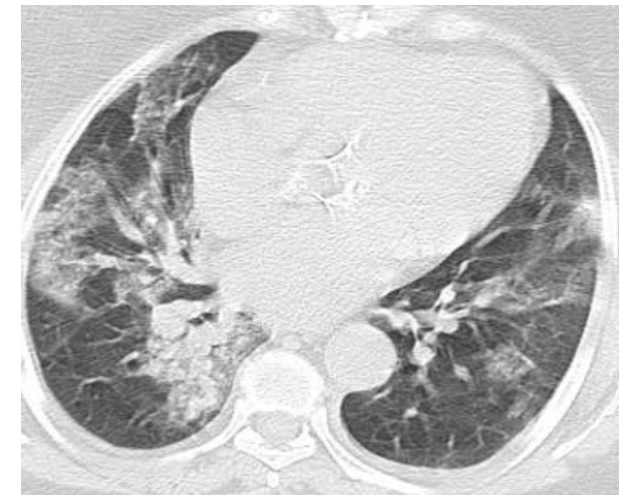
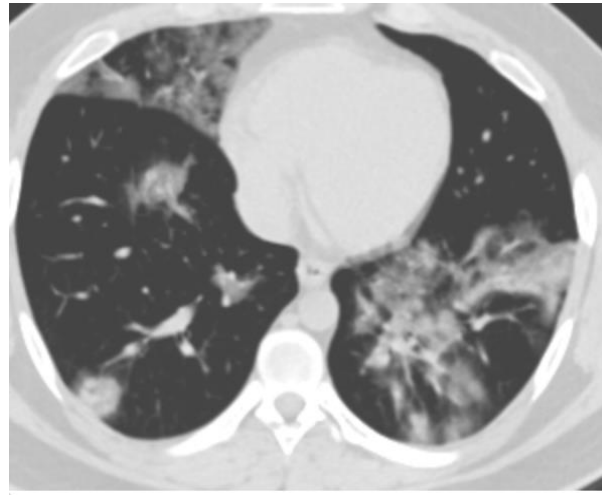
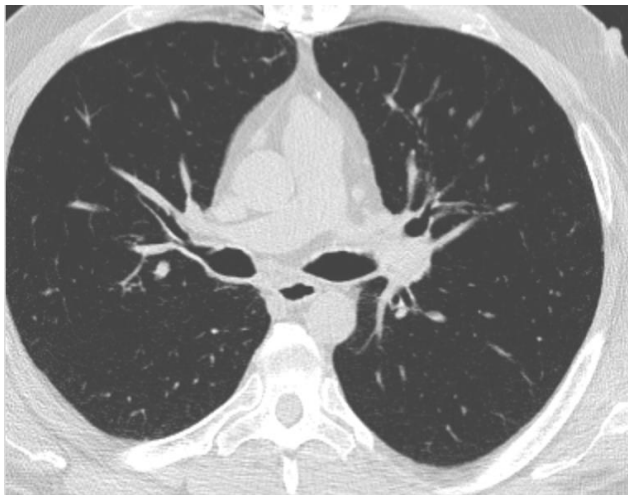
Axial CT Sections:

- (A) Ground-glass opacities
- (B) Consolidation
- (C) Crazy-paving
- (D) Sub-pleural linear opacities and parenchymal bands
- (E) Halo sign
- (F) Reverse halo sign

Garg, M., et. al. (2021) The Indian journal of medical research.

Publicly available SARS-CoV-2 CT scan dataset

- **1252** CT scans for positive SARS-CoV-2 infection (COVID-19)
- **1230** CT scans for non-infected patients
- **2482** CT scans in total.

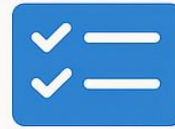


Benefits of Deep Learning for COVID-19 Diagnosis

Benefits of Deep Learning to COVID-19 Diagnosis



Rapid Automated
Screening



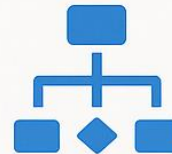
Improved Consistency
& Reproducibility



Quantitative
Monitoring



Enhanced Sensitivity
& Specificity



Triage & Resource
Optimization

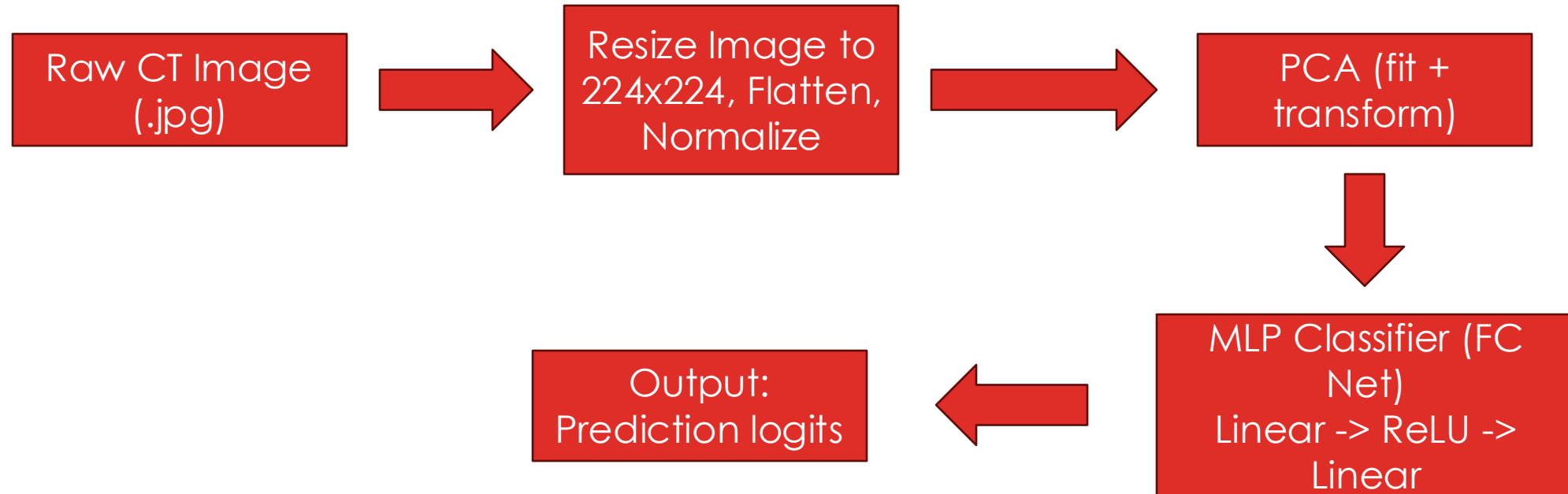


Reduction of Workload
& Exposure



Adaptability & Continuous Learning

Methodology and Workflow



PCA and SVD

PCA

- Finds orthogonal basis vectors of maximum variance
- Projects data into lower-dimension space
- Generally defined via eigen-decomposition of the covariance matrix

SVD

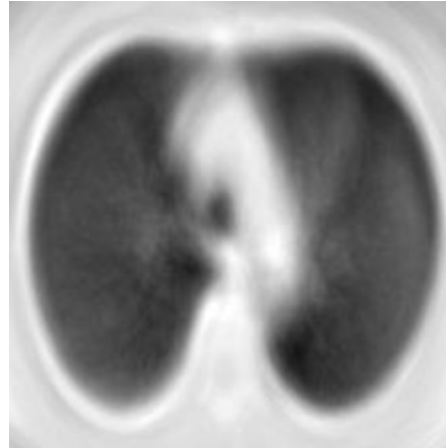
- $A = U\Sigma V^T$
- $A^T A = (U\Sigma V^T)^T (U\Sigma V^T)$
- $A^T A = V\Sigma^T U^T U\Sigma V^T$
- $A^T A = V\Sigma^T \Sigma V^T$; $D = \Sigma^T \Sigma$
- $A^T A = VDV^T$
- Diagonalization of a matrix
- D are the eigen values of $A^T A$
- V are the eigenvectors
- $A^T A$ is our covariance matrix, still needs to be normalized

Eigen-Images

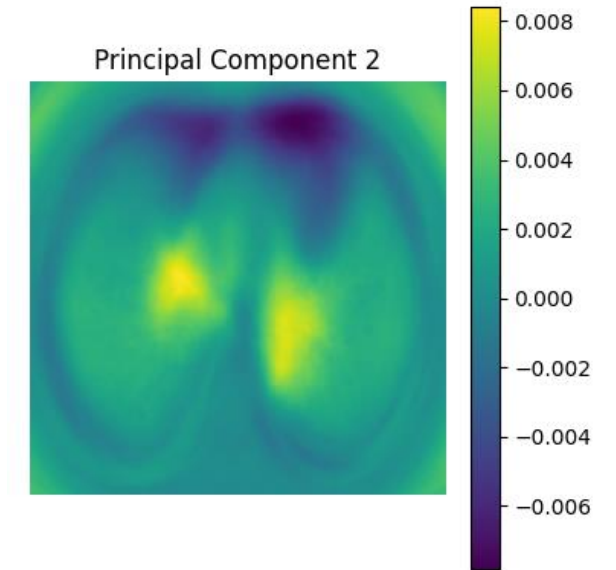
Original COVID
Lung



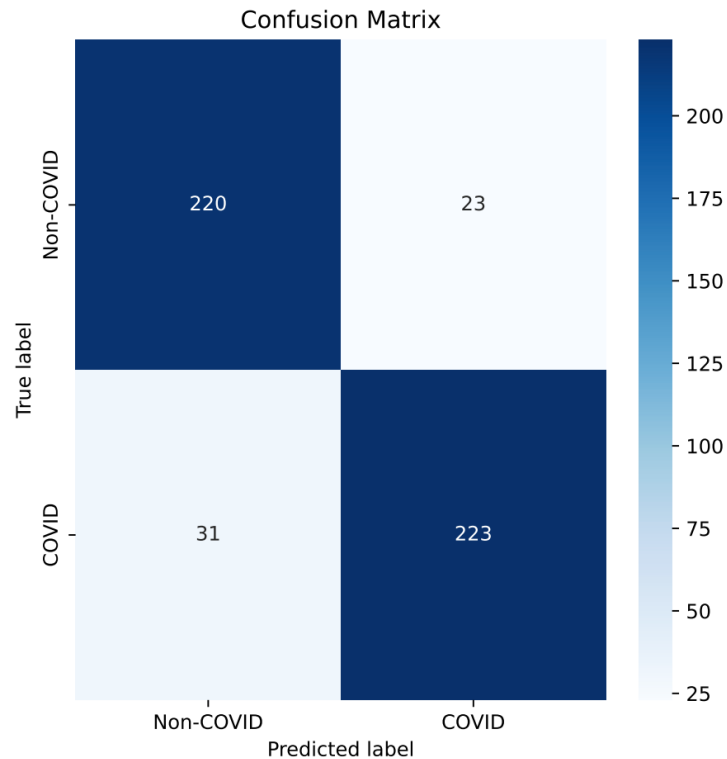
PCA'ed COVID
Lung



Heatmap of Principal
Component 2
(Important to model)

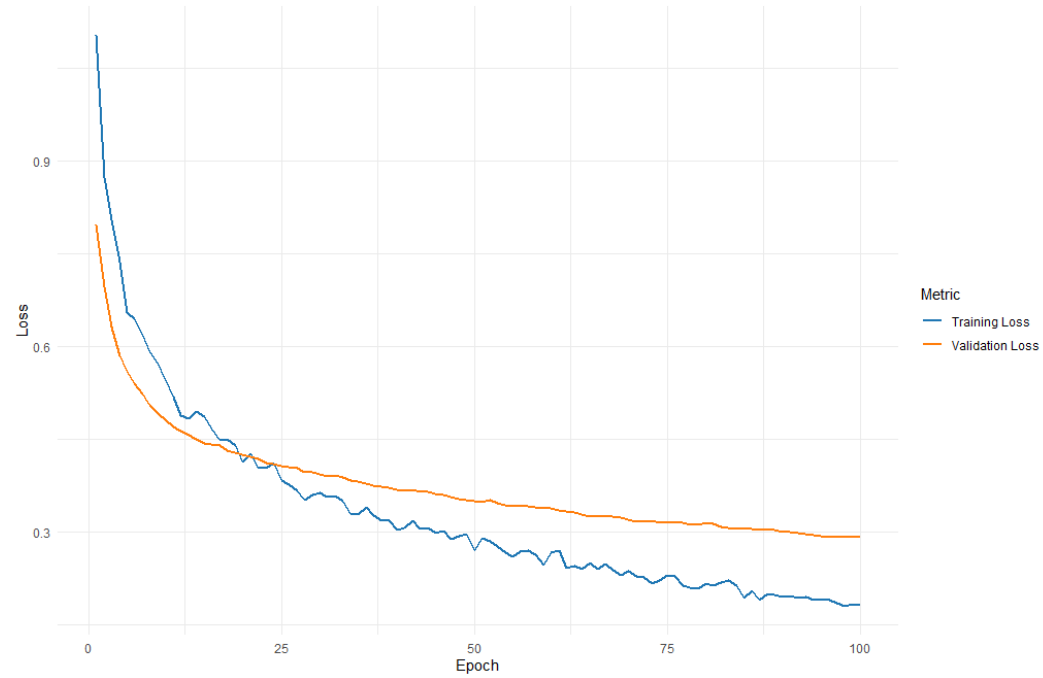


- **Very Performant**
- **High Precision**

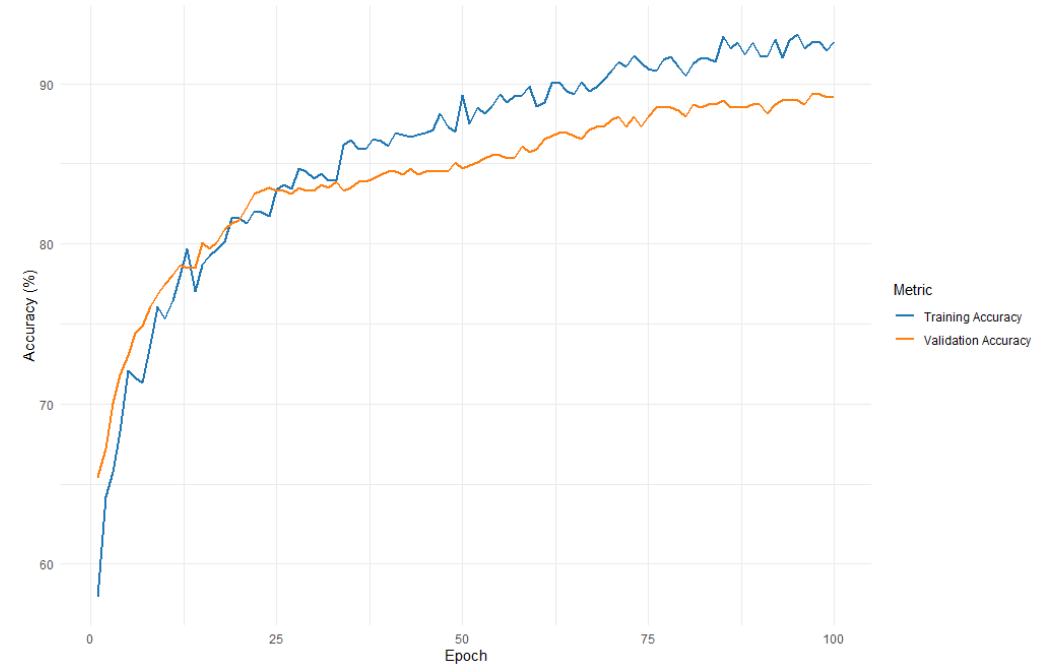


Metric	Value
Accuracy	89.11%
Precision	90.65%
Recall	87.80%
F1 Score	89.20%

Training vs Validation Loss



Training vs Validation Accuracy



Conclusion and Future Directions

- Overall, this work is a promising start towards a full-fledged model for pulmonary infections
- Future work would focus on increasing explainability for doctors
 - Perhaps, a composite of the most important PCs or SHAP
 - Ensemble approach using a traditional CNN for Grad-CAM?
- Future work could also try nonlinear dimensionality reduction such as autoencoders or UMAP, in case model has nonlinear relation

ACKNOWLEDGEMENT

Department of Mathematics, University of
Wisconsin in Madison, Madison, WI, USA

Thank you
for
listening!



Asst. Prof. Dr. Pouria Behnoudfar, Ph.D.